SWEN30006
Software Modelling and Design

GRASP:
More Objects with Responsibilities

Textbook: Larman Chapter 25

“Luck is the residue of design."

—Branch Rickey

Learning Objectives

On completion of this topic, you should be able to:

- Apply four of the GRASP principles or patterns for Object-Oriented Design.
  - Indirection
  - Pure Fabrication
  - Polymorphism
  - Protected Variation

(Revisited) GRASP: General Responsibility Assignment Software Patterns/Principles

- **GRASP – Definition:** A set of patterns (or principles) of assigning responsibilities into an object.
- GRASP aids in applying design reasoning in a methodical, rational, and explainable way
  - Given a specific category of problem, GRASP guides the assignment of responsibilities to objects
- Useful to know and support Responsibility-Driven Design (RDD)
- **Pattern – Definition:**
  - A *named* and *well-known* problem/solution pair that can be applied in new contexts
  - A recurring successful application of expertise in a particular domain

(Last Week) GRASP: Learning Aid for RDD

- This subject will cover the 9 GRASP patterns/principles
  - Creator
  - Information Expert
  - Low Coupling
  - High Cohesion
  - Controller*
  - Polymorphism
  - Indirection
  - Pure Fabrication
  - Protected Variations

*Controller will be covered in Workshop supplementary material

(This Week) GRASP: Learning Aid for RDD

- This subject will cover the 9 GRASP patterns/principles
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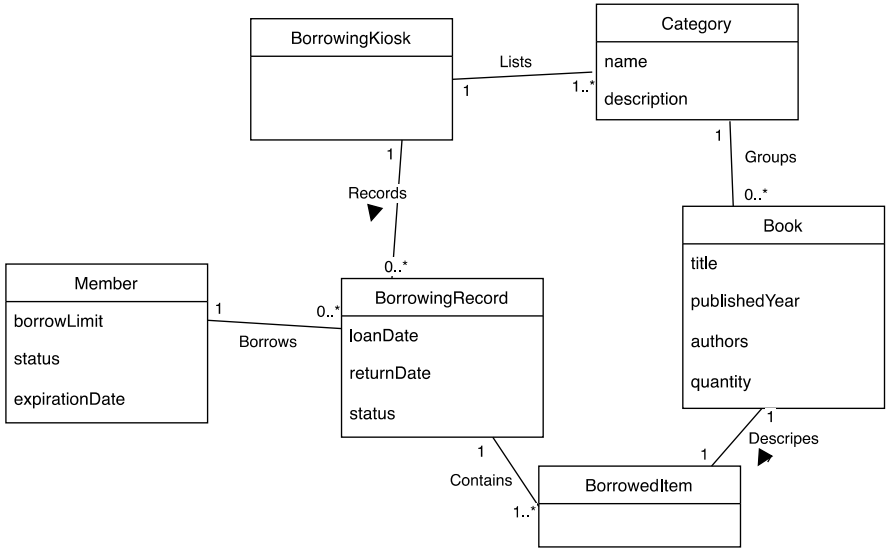
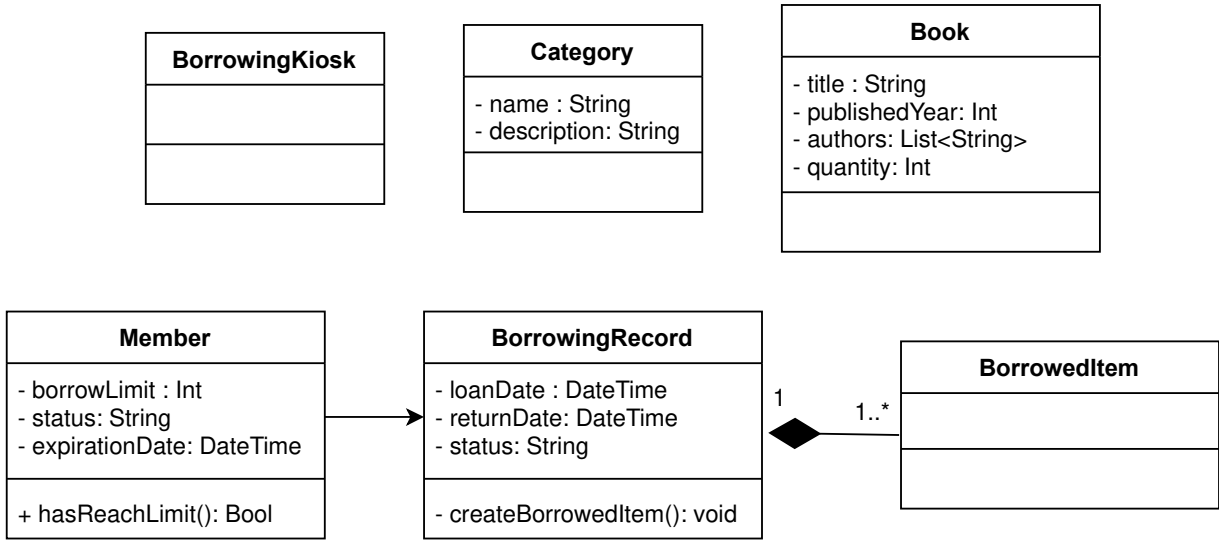
(Last Week) Example: Designing the design model based on the domain model



A **domain model** of
Library Borrowing System

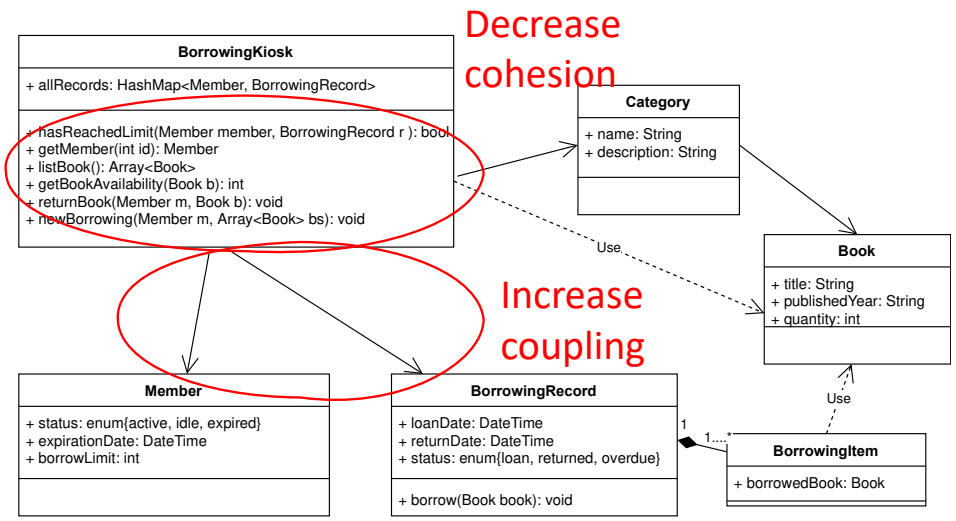


An *incomplete* **design model** of
Library Borrowing System

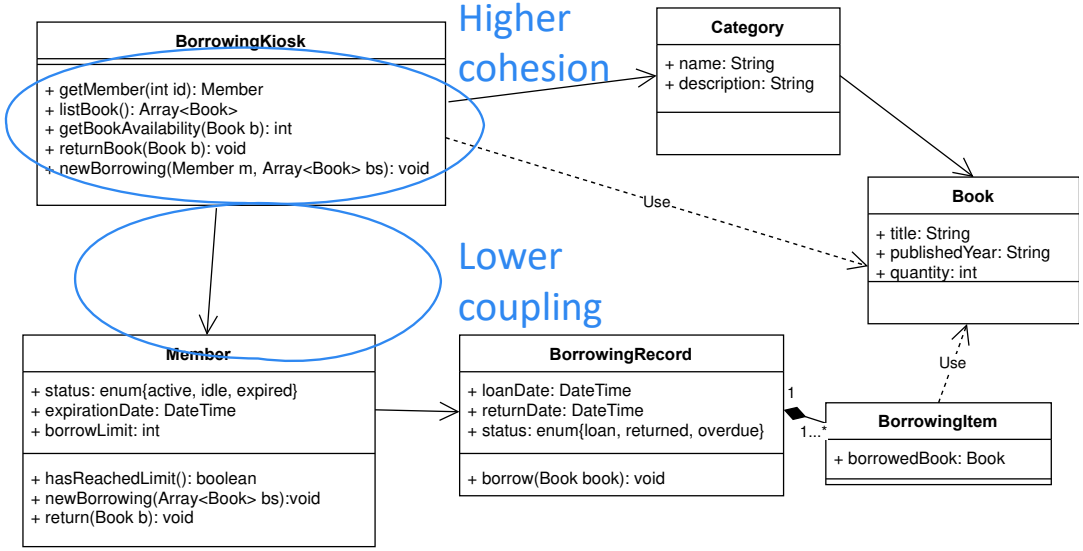
Why can't the BorrowingKiosk class have responsibility of "hasReachedLimit" instead of the Member class?

(Last Week) Example: Designing the design model based on the domain model

A **design model** of Library Borrowing System
Design option 1:
Assign “hasReachedLimit” to the **BorrowingKiosk** class



A **design model** of Library Borrowing System
Design option 2:
Assign “hasReachedLimit” to the **Member** class

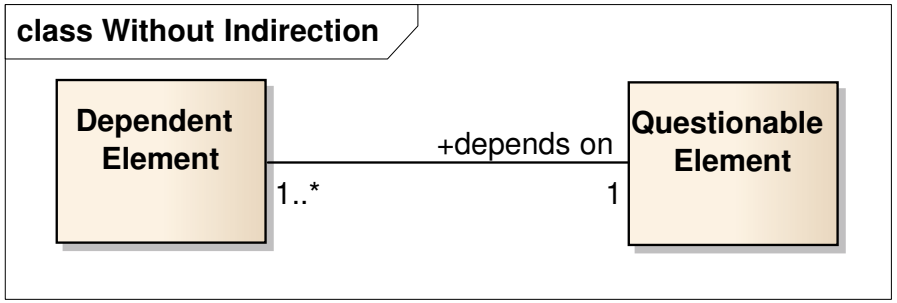


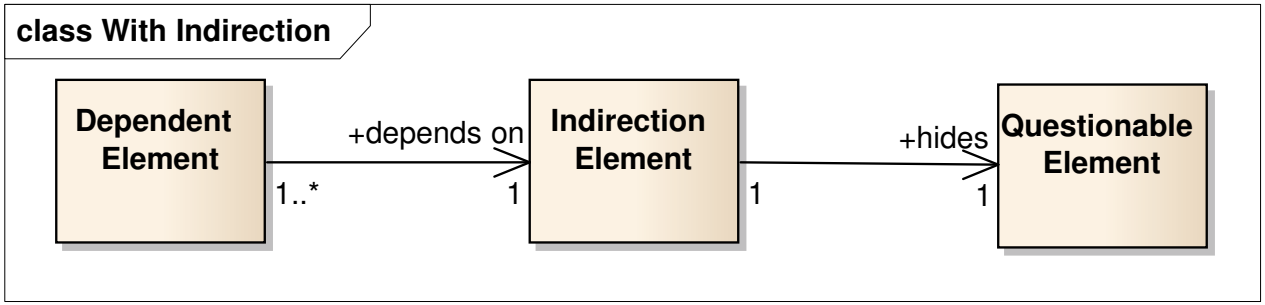
Using GRASP’s **Indirection** pattern

GRASP: Indirection

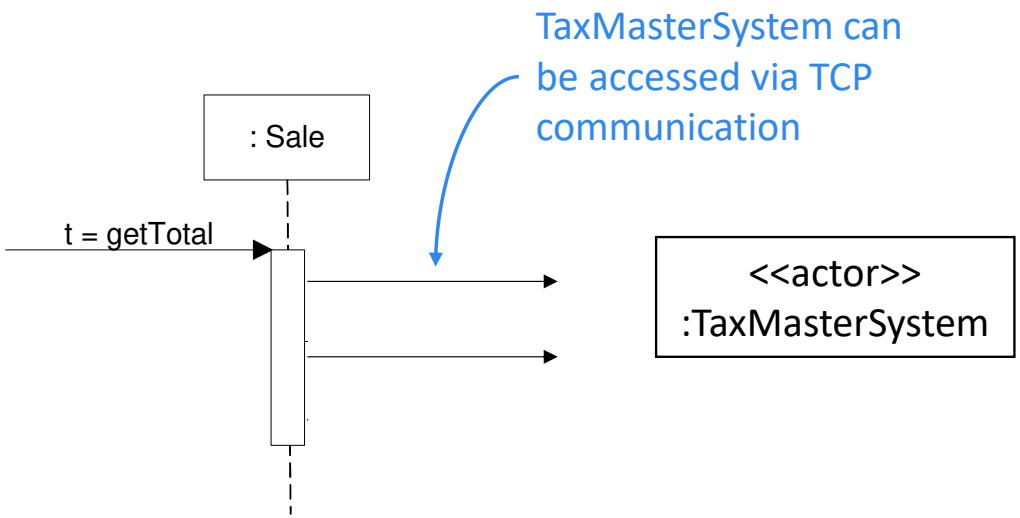
Indirection

- **Problem:** Where to assign a responsibility, to *avoid direct coupling* between two (or more) things?
  - How to de-couple objects so that low coupling is supported, and reuse potential remains higher?
- **Solution:** Assign the responsibility to an *<u>intermediate</u>* object to mediate between other components or services so that they are not directly coupled.
  - The intermediary creates an *indirection* between the other components.





Example: POS



POS will use Tax Master System (an external service) to calculate tax. Who should be responsible for calculating tax?

- By Expert, Sale is responsible for calculating total. So, Sale should also calculate tax
- Should Sale directly use TaxMasterSystem via TCP communication?

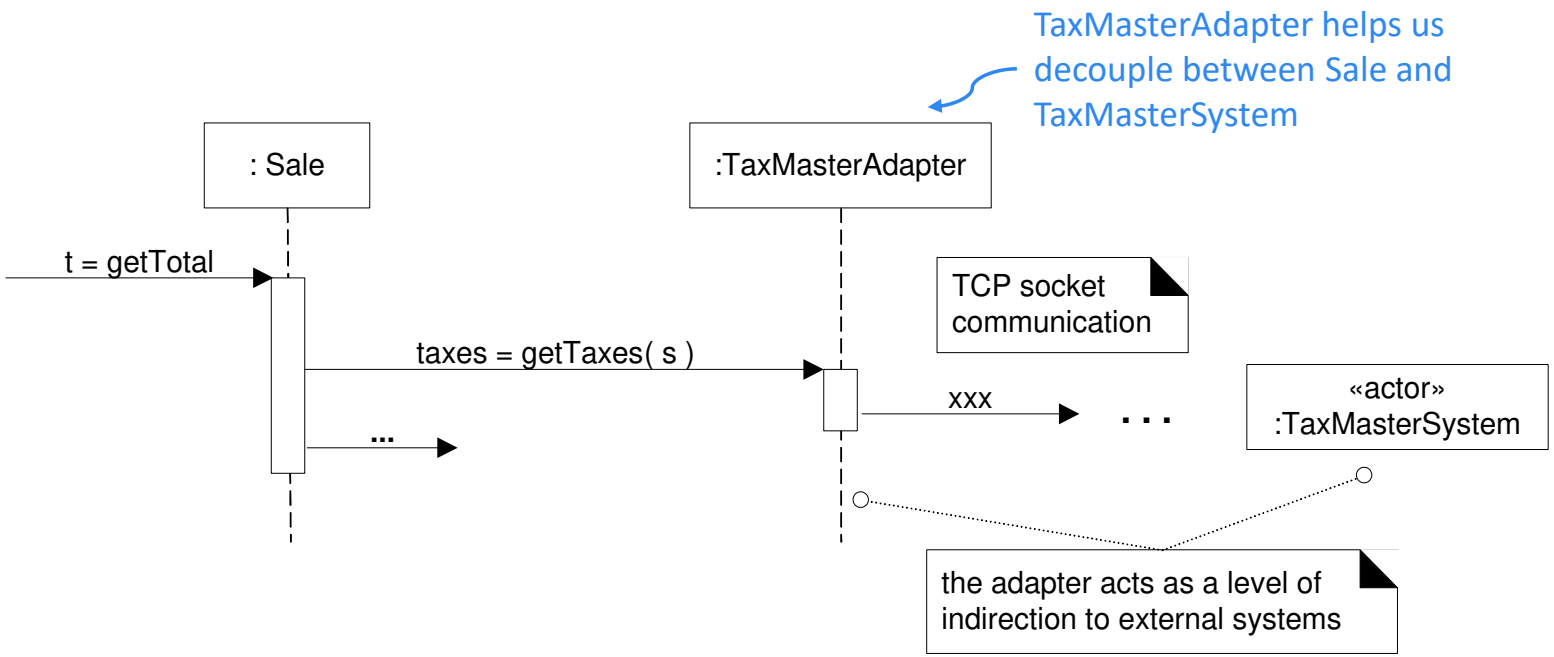
Discussion: If this responsibility is assigned to Sale,

→ a lot of TCP communication to TaxMasterSystem will be implemented in Sale

→ Sale class is highly coupled with TaxMasterSystem

Example: POS

Solution: By Indirection, the responsibility of communicating with TaxMasterSystem should be assigned to an intermediate class, i.e., TaxMasterAdapter



(Last Week) Example: Designing the design model based on the domain model

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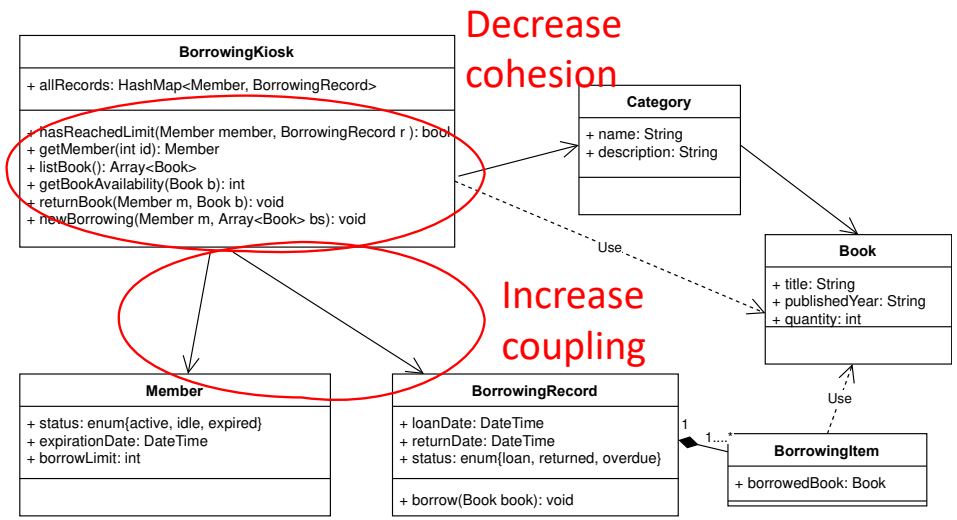

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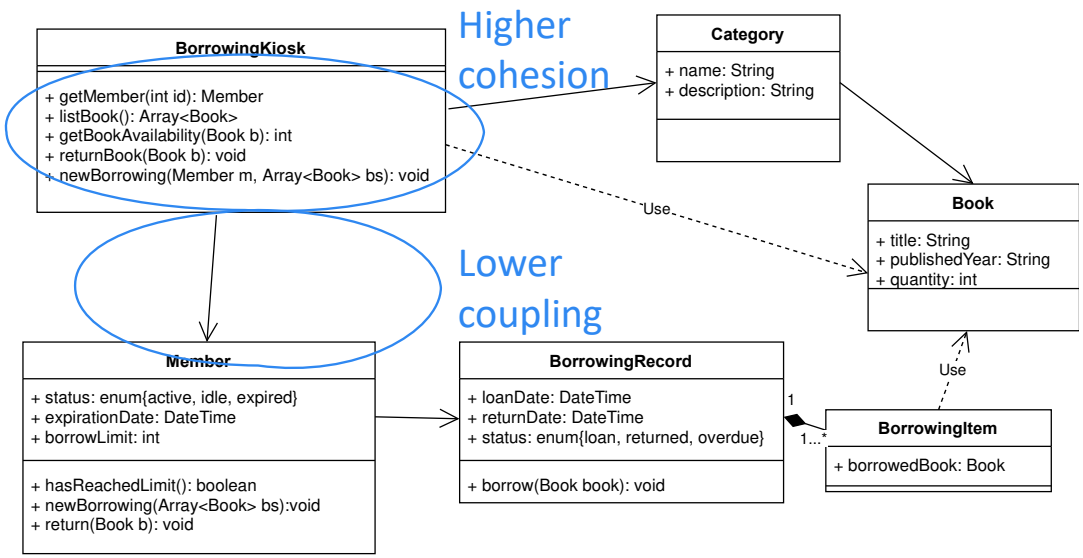


To improve, we can lose coupling by assigning Member **as an intermediate class** between BorrowingKiosk and BorrowingRecord

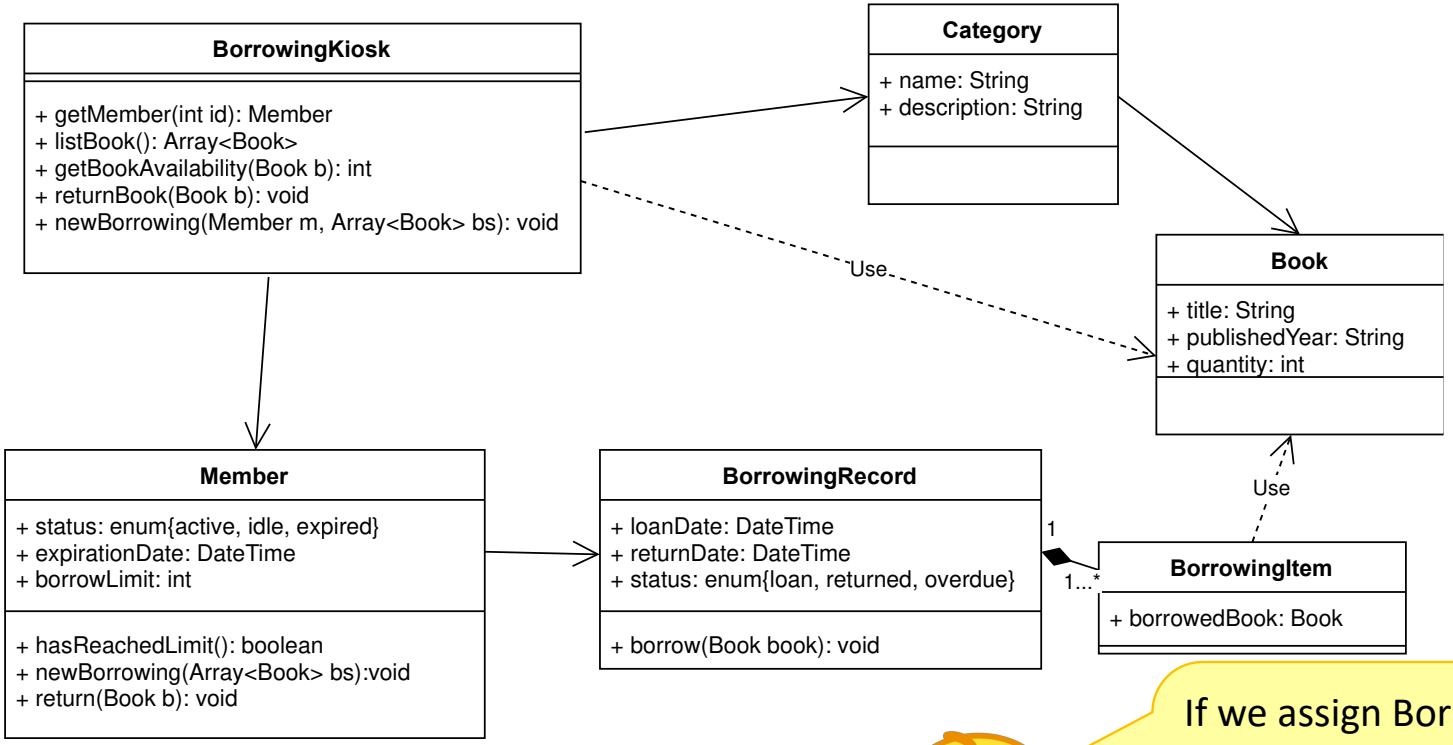
Indirection: Discussion

- The motivation for Indirection is usually Low Coupling (i.e., decouple objects for future reuse)
- Old adage:

  "Most problems in computer science can be solved by another level of indirection"
- Counter adage:

  "Most problems in performance can be solved by removing another layer of indirection!"

Example: Designing the design model based on the domain model



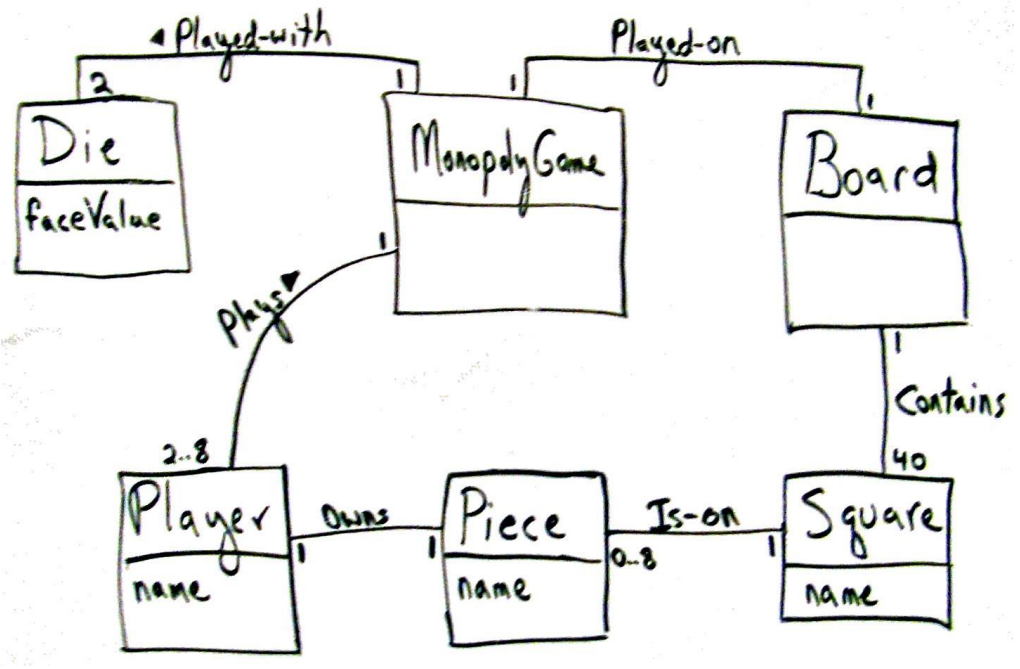
If we assign BorrowingKiosk to be responsible for getting user inputs from UI (i.e., a controller), who should be responsible for application logic (e.g., newBorrowing(), returnItem(), getItemAvailability())

GRASP:
Pure Fabrication

Pure Fabrication

- **Problem:** Which object should have responsibility when you do not want to violate *High Cohesion* and *Low Coupling*, but solutions offered by other patterns (e.g., Expert) are not appropriate?
  - Domain model often inspires the design of software objects (achieving low representational gap)
  - In many situations, assigning responsibilities only based on the domain model often leads to the problem of poor cohesion or coupling.
- **Solution:** Assign a highly cohesive set of responsibilities to an *artificial* or convenience class that does not represent a problem domain concept
  - Make up a class to support high cohesion, low coupling, and reuse

Example: Monopoly

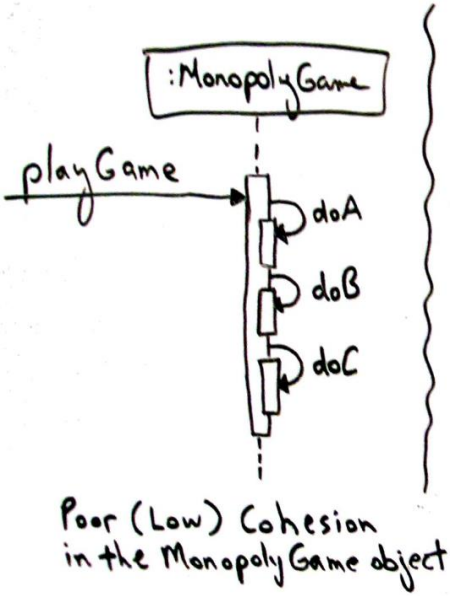


Monopoly Game Simulation
Domain Model

Who should be responsible for rolling the dice?

- Based on the domain model & Expert, MonopolyGame class should be assigned for this responsibility

<u>Discussion:</u> It is acceptable, but leads to low cohesion



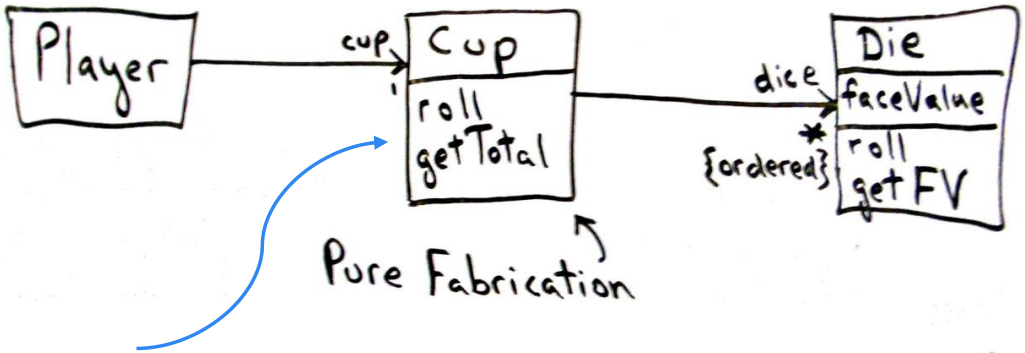
Based on the domain model, the design can violate high cohesion

Let's use **Pure Fabrication**!

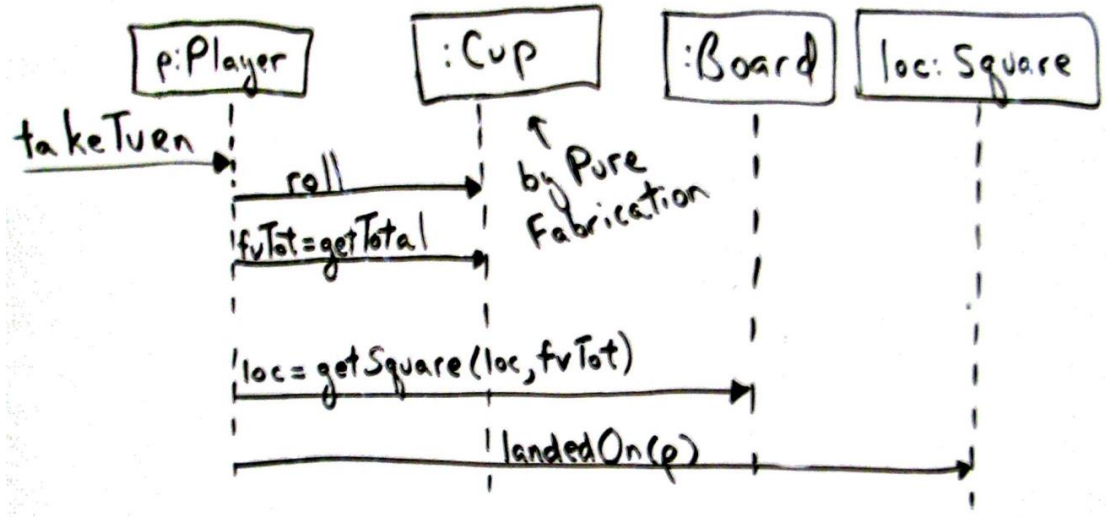
Example: Monopoly

Who should be responsible for rolling the dice?

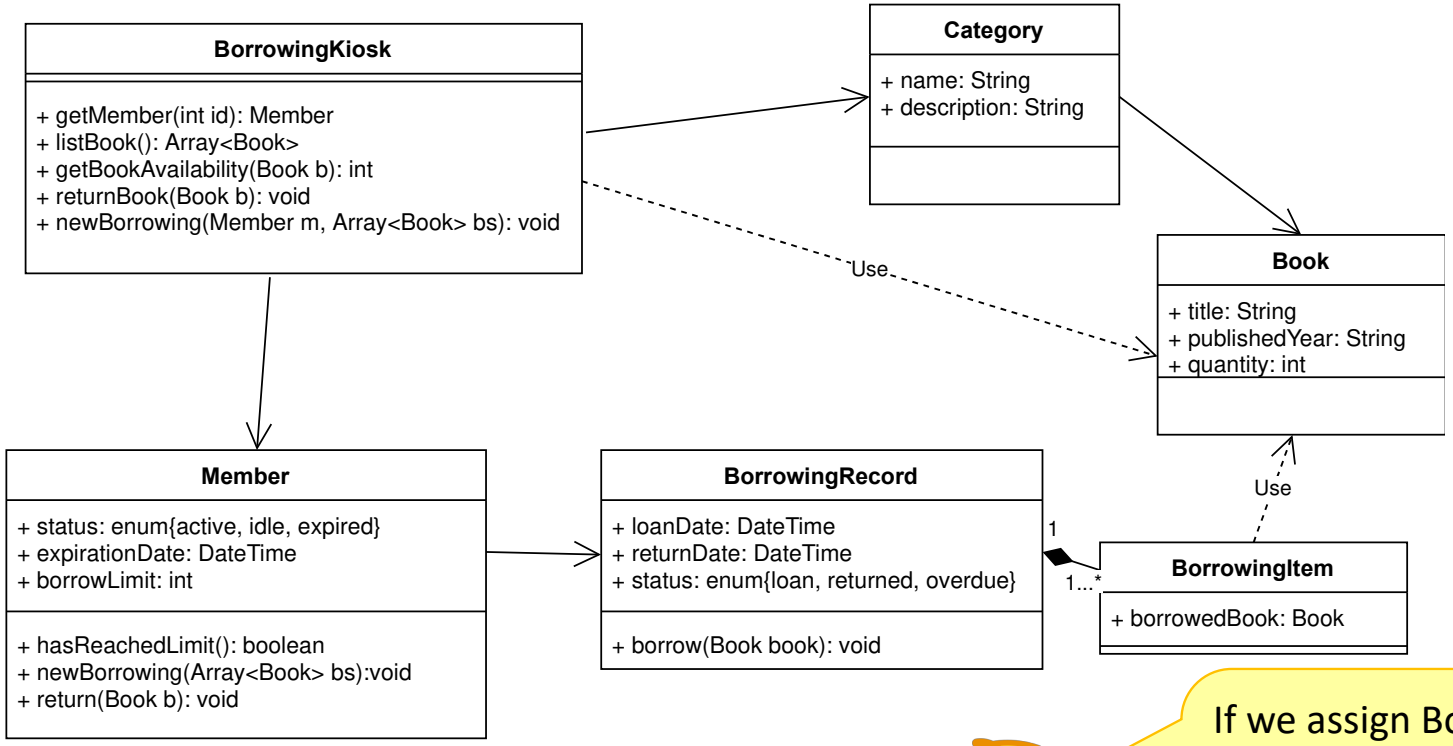
<u>Solution</u>: By Pure Fabrication, let's create an artificial class which is responsible for rolling dice



In the domain, Cup was not used in Monopoly Game



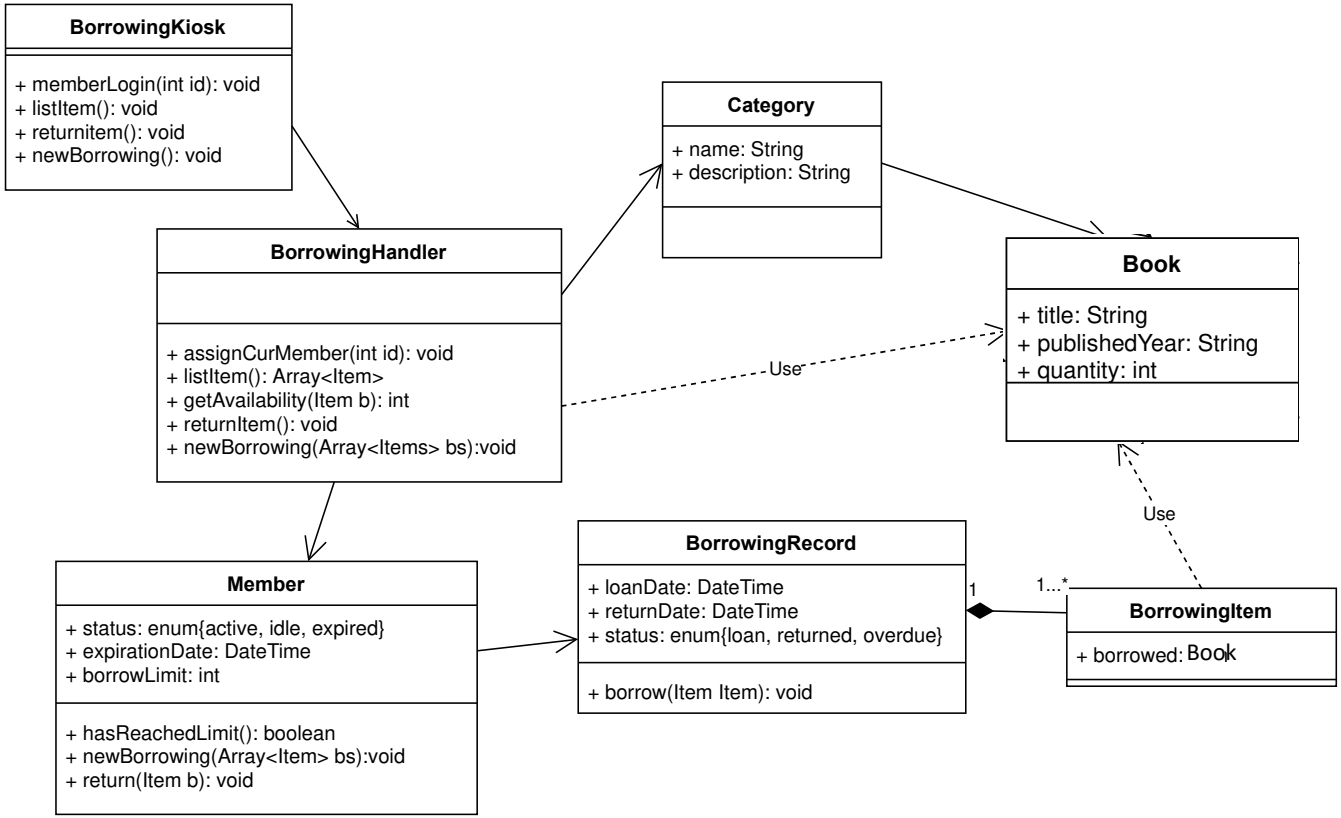
Example: Designing the design model based on the domain model



Keeping the **High Cohesion and Low Coupling** principles in mind, *none* of the existing classes are suitable!

If we assign BorrowingKiosk to be responsible for getting user inputs from UI (i.e., a controller), who should be responsible for application logic (e.g., newBorrowing(), returnItem(), getItemAvailability())

Example: Designing the design model based on the domain model



To achieve high cohesion, i.e., let BorrowingKiosk focus on handling user inputs, we should create a new class to handle the items and borrowing (Let's called BorrowingHandler).

Pure Fabrication: Contraindications

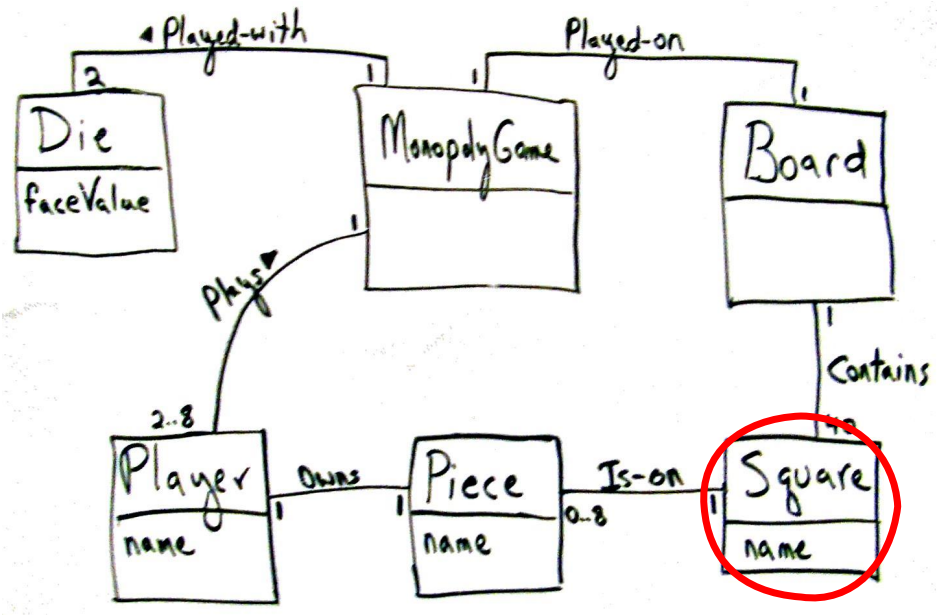
- Many Pure Fabrications are generated because of Indirection.
  - Indirection: Assigning any class as an intermediary
  - Pure Fabrications: Assign an artificial class (that does not represent the domain) as an intermediary
- Pure Fabrication should *not* be used as excuse to add new objects
- If overused Pure Fabrication, the design can end up with one class has one responsibility
  - Too many behaviour objects that have responsibilities *not* co-located with the information required for their fulfillment
  - can adversely affect coupling (i.e., high dependencies)

GRASP: Polymorphism

Polymorphism

- **Problem (1):** How to handle alternatives based on type?
  - Conditional variation: Adding new alternatives and if-then-else (or case-statement) are required many places
  - E.g., behaviour varies based on some conditions
- **Problem (2):** How to create pluggable software components?
  - E.g., Client-Server relationship: How to replace Server component without affecting Client?
- **Solution:** When related alternatives or behaviours vary by type (class), assign responsibility for the behaviour—using polymorphic operations—to the types for which the behaviour varies
  - Polymorphic operations: Operations with the same interface
  - ***Corollary*:** Do not test for the type of an object and use conditional logic to perform varying alternatives based on type.

Example: Monopoly



Monopoly Game Simulation Domain Model

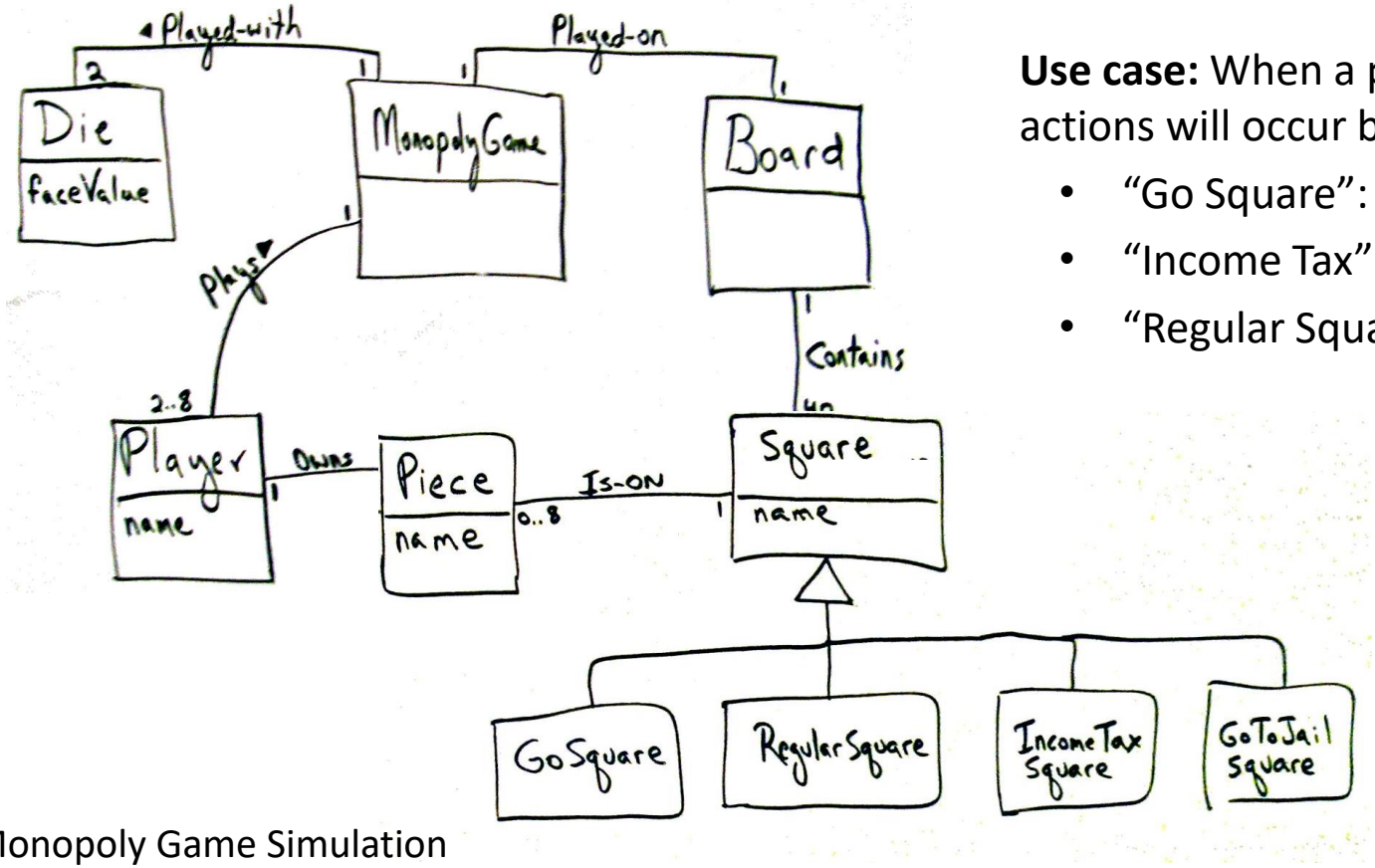
Use case: When a player lands on a square, different actions will occur based on the types of square, e.g.,

- "Go Square": A player receives $200
- "Income Tax": A player pays a tax of the income
- "Regular Square": A player does nothing

<u>Realization:</u> The different types of Square can be seen as concepts in the domain as well. It should be captured in the *domain* model

→ Since the concepts are similar, use a **generalization specialization class hierarchy (class hierarchy)** to organize these concepts

Example: Monopoly – Update *Domain* Model



Monopoly Game Simulation Domain Model

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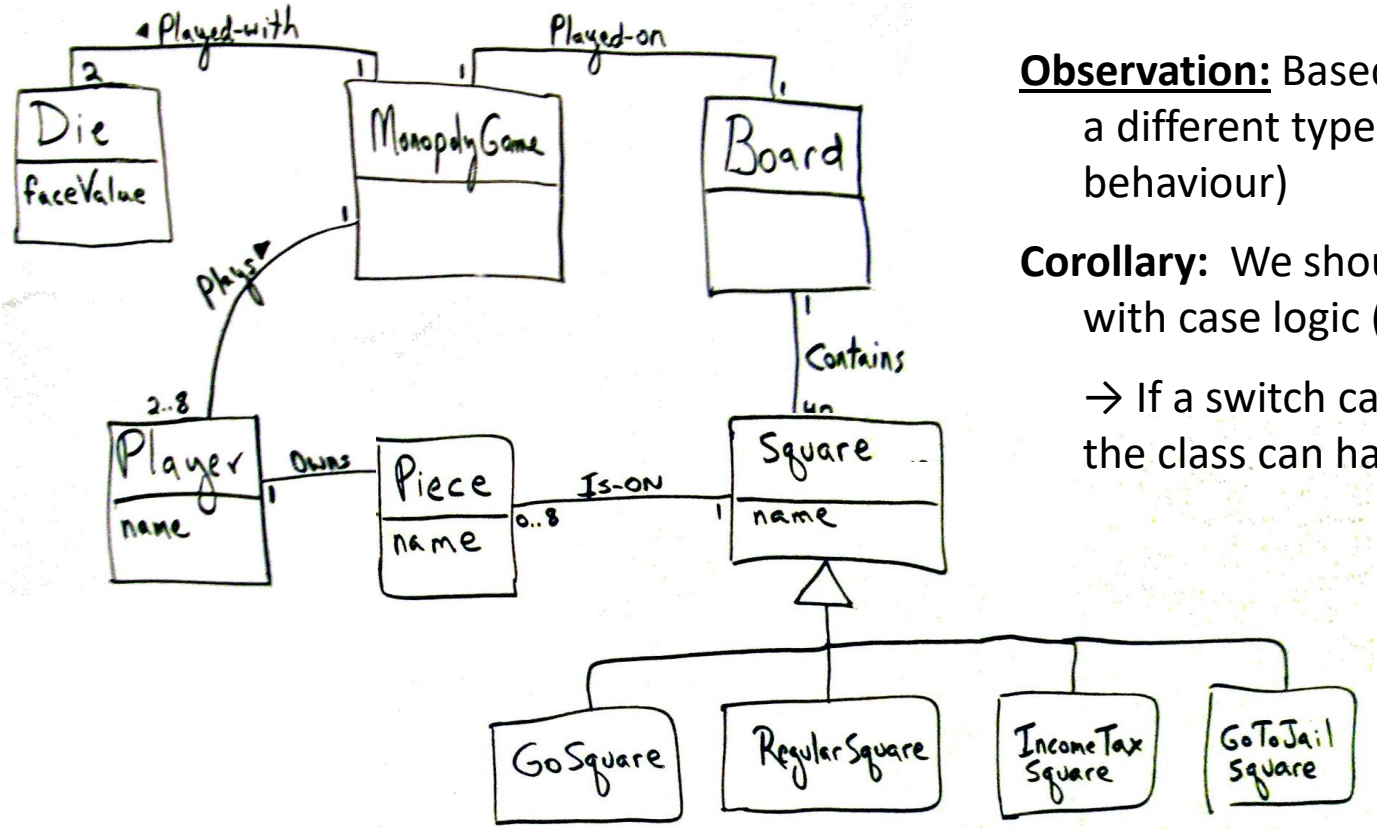
Example: Monopoly – Create *Design* Model (1)



Monopoly Game Simulation Domain Model

<u>Observation:</u> Based on Use case and domain model, a different type of square has a different rule (i.e., behaviour)

Corollary: We should not design our software objects with case logic (e.g., a switch statement)

→ If a switch case (or if-else) is used in Square class, the class can have *low cohesion* and can be bloated!

Example: Monopoly – Create *Design* Model (2)

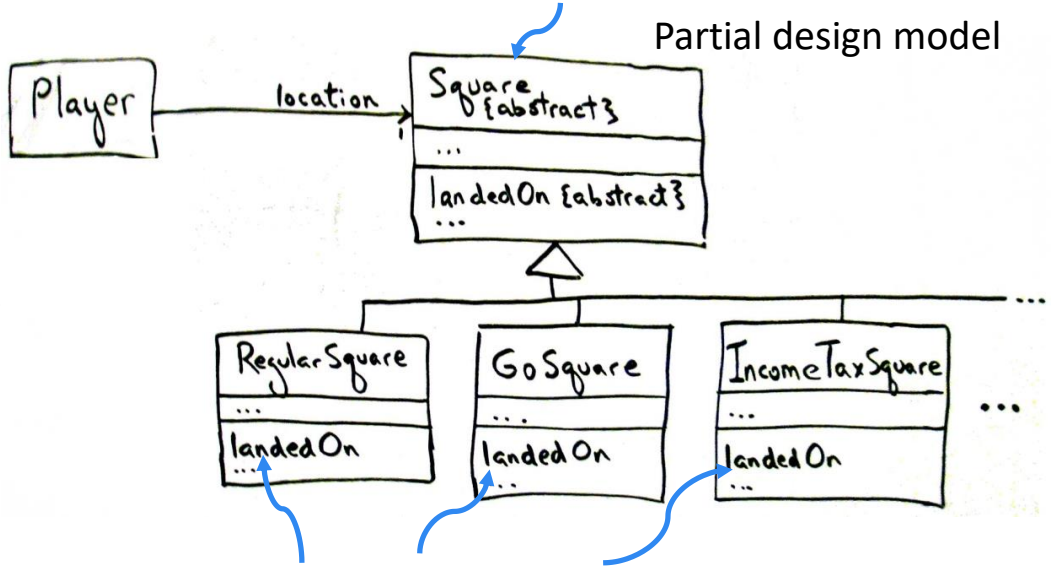
Analysis: There are alternatives based on type of Squares
→ Polymorphism should be used.

Approach: Based on Polymorphism, we should create a polymorphic operation for each type for which the behavior varies.

- What are types that Square varies?
  → ”Go Square”, “RegularSquare”, “IncomeTaxSquare”, so on
- What is the operation that varies?
  → The behaviour of Square will vary when a player *lands on* a square
  → Then, the responsibility of “landedOn” should be a *polymorphic operation*

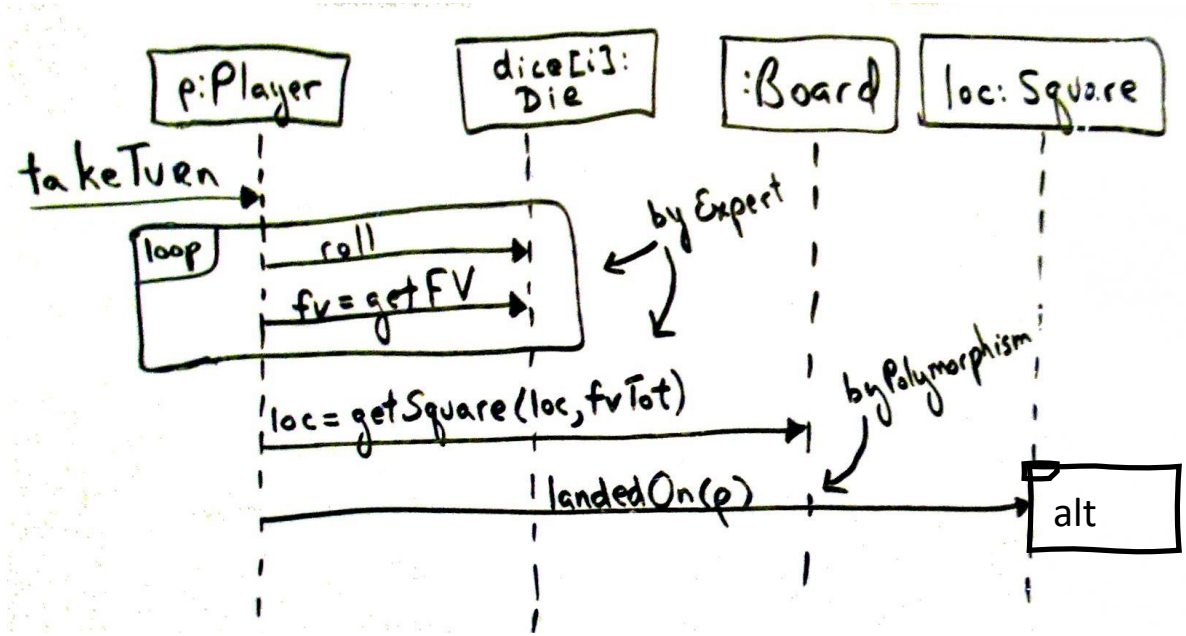
Use an abstract class to create a common interface

Partial design model

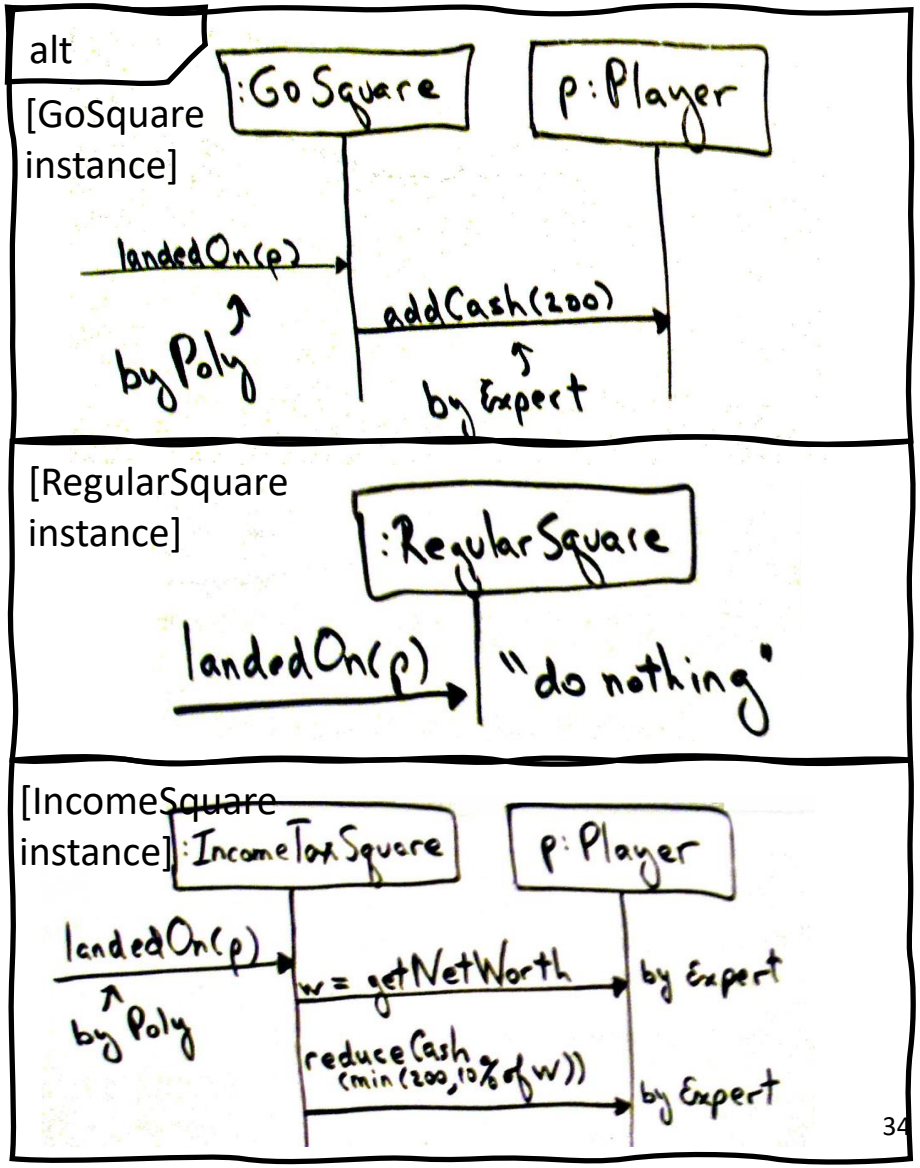


Implements landedOn operation based on type

Example: Monopoly



The behaviour of square is based on the type of Square that Board returns to Player

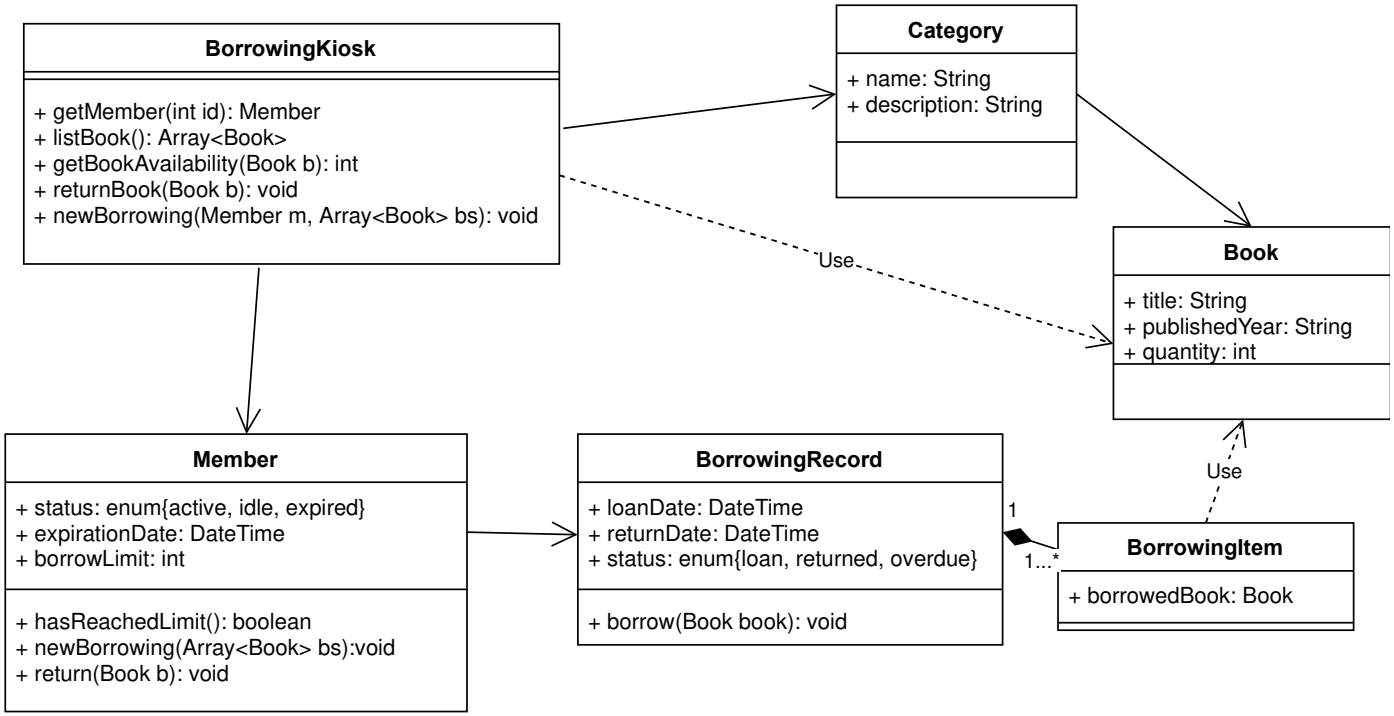


sub-diagram

Polymorphism: Discussion

- Polymorphism is a *fundamental* pattern in designing how a system is organised to *handle similar variations*
  - Based on Polymorphism, a design is easily extended to handle new variations
- **Contraindications**:
  - Designing systems with interfaces and polymorphism for speculative "future-proofing" against an unknown variation could be useful.
  - However, the unnecessary effort might occur if polymorphism is applied at the points where variations, in fact, are improbable and will never actually arise.

Example: Designing the design model based on the domain model



What if in the future, a member can borrow something else other than a book (e.g., a video) via Kiosk as well. Other items might have different attributes.

GRASP:
Protected Variations

Protected Variation

- **Problem:** How to design objects, subsystems, and systems so that the variations or instability in these elements does not have an undesirable impact on other elements?
  - Points of change include:

    Variation point: variations in existing system or requirements

    Evolution point: speculative variations that may arise in the future
- **Solution:**
  - Identify points of known or predicted variation or instability;
  - Assign responsibilities to create a stable interface* around them

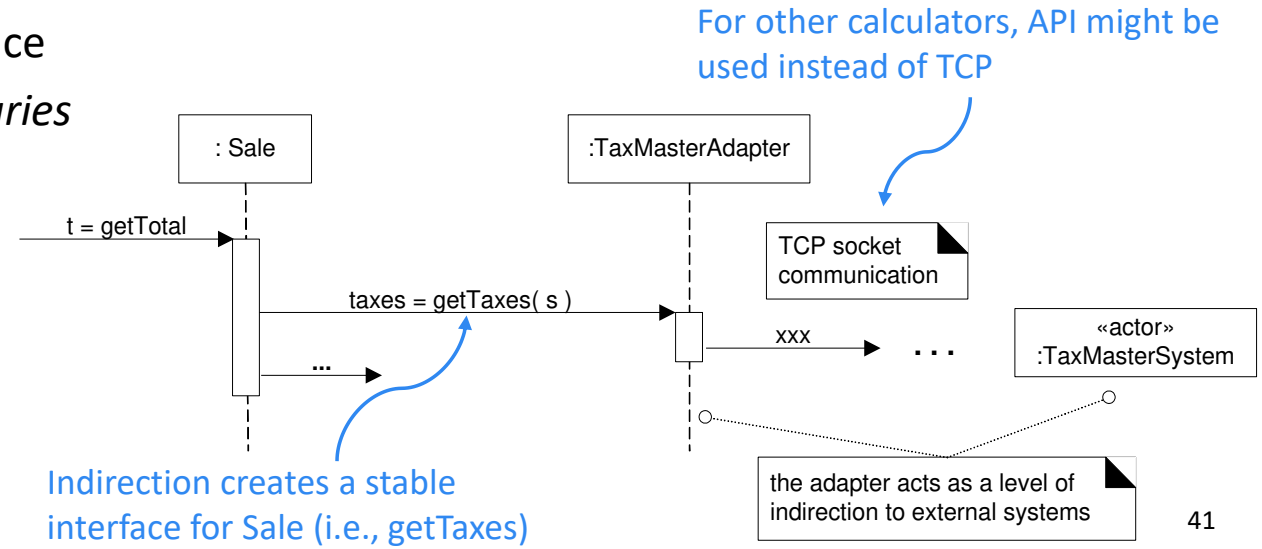
**Interface:* a means of access (not only a programming language interface or Java interface)

Example: POS

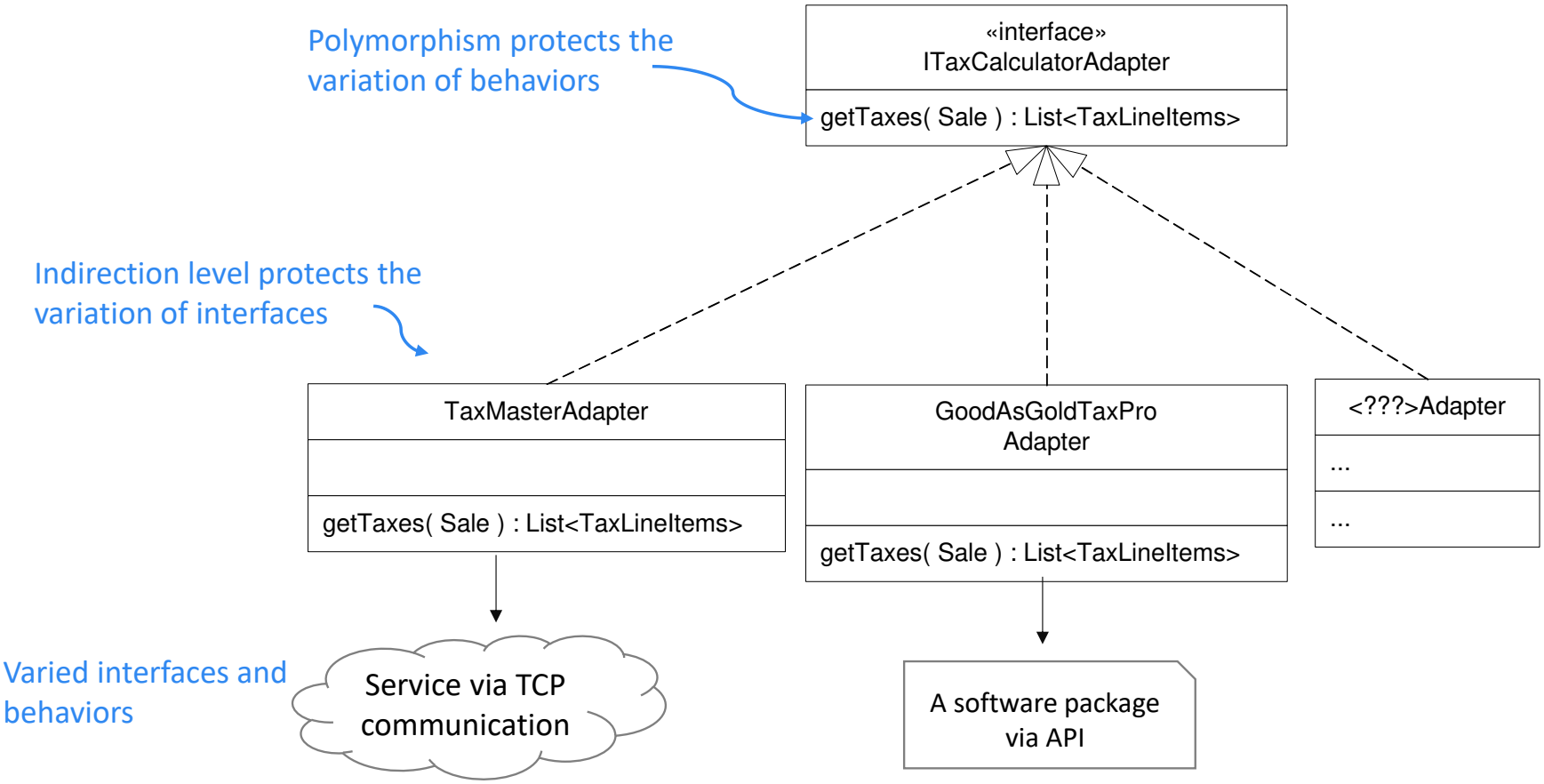
POS currently uses 'TaxMasterSystem' to calculate tax. In the future, other tax calculators might be used.

Analysis: For protected variation:

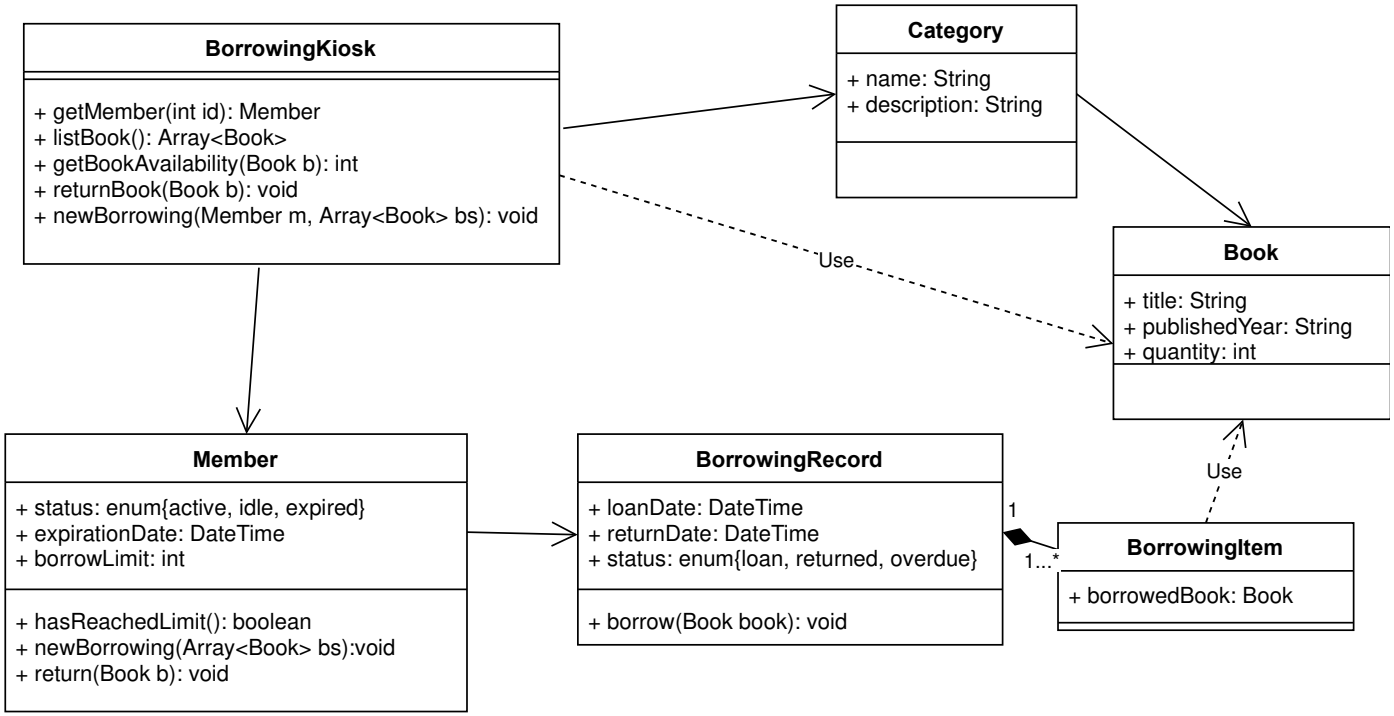
1. What is the point that will be known or predicted variation or instability?
   → The point of instability or variation is the different interfaces or APIs of external tax calculators
2. Assign responsibilities to create a stable interface* around them
   → Use Indirection to create a stable interface
   → Note that the behaviour of getTaxes() *varies* based on the type of the tax calculator
   → By Polymorphism, let's make getTaxes() a polymorphic operation!



Example: POS


Polymorphism protects the variation of behaviors
«interface»
ITaxCalculatorAdapter
getTaxes(Sale) : List<TaxLineItems>
Indirection level protects the variation of interfaces
TaxMasterAdapter
getTaxes(Sale) : List<TaxLineItems>
GoodAsGoldTaxPro Adapter
getTaxes(Sale) : List<TaxLineItems>
<???>Adapter
...
...
Varied interfaces and behaviors
Service via TCP communication
A software package via API

Example: Designing the design model based on the domain model



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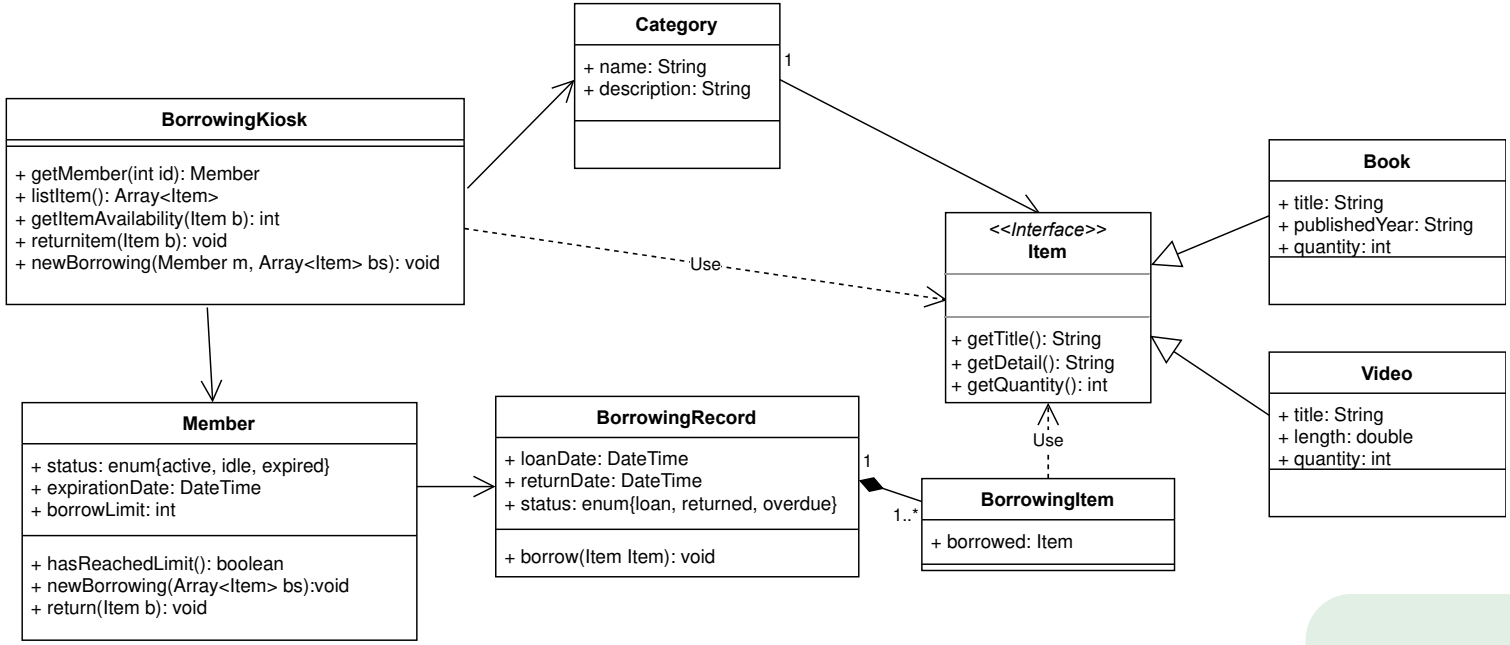
Example: Designing the design model based on the domain model



To **protect future variation** of items, we use **Polymorphism** on Item, e.g., make getDetail() a polymorphic operation

Protected Variation: Discussion

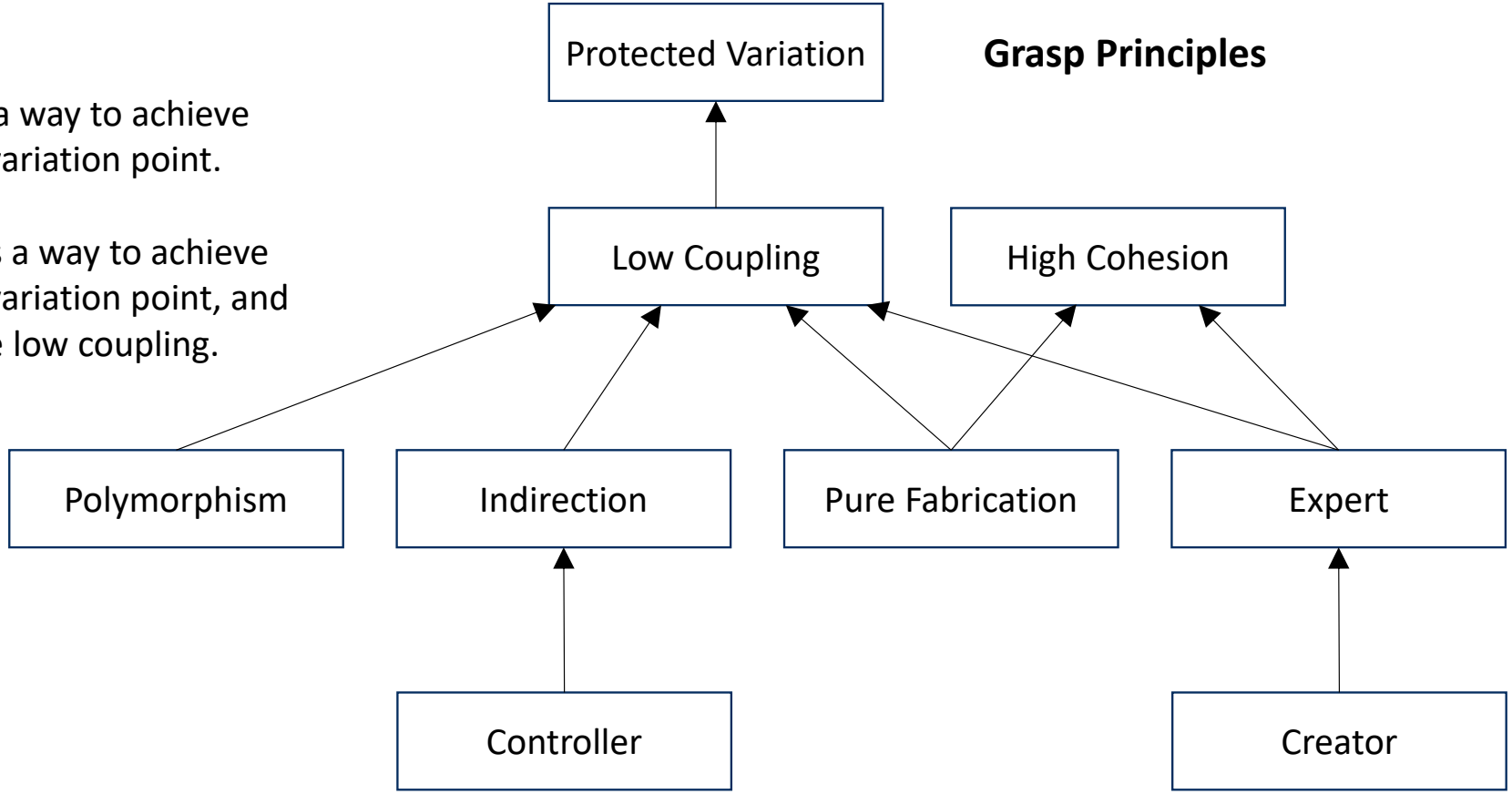
- Protected Variation (PV) is *very* important, fundamental <u>principle</u> of software design!
- PV is a root principle motivating most of the mechanisms and patterns in programming and design.
  - For example: Core Protected Variations Mechanisms, Data-Driven Design, Service Lookup, Interpreter-Driven Design, (See more in textbook)
- PV is equivalent to **Open-Closed Principle** (OCP), described by Bertrand Meyer
  - OCP: Modules should be both open (for extension; adaptable) and closed (for modification that affects clients).
  - A class can be closed w.r.t. attribute access (access methods only – no changes), but open to modification of underlying attributes (+ addition of new methods).
- **Contraindications**: Similar to Polymorphism
  - Cost of speculative "future-proofing" at evolution points can outweigh the benefits
  - It may be cheaper/easier to rework a simple "brittle" design

Relationship between GRASP Principles

For example:

Low coupling is a way to achieve protection at a variation point.

Polymorphism is a way to achieve protection at a variation point, and a way to achieve low coupling.



Summary & Remarks

- Protected Variation, Low Coupling, and High Cohesion are **fundamental principles** that we aim to achieve when designing software objects
- Use the remaining 6 GRASP patterns to achieve the principles
  - Creator
  - Information Expert
  - Controller
  - Polymorphism
  - Indirection
  - Pure Fabrication

Lecture Identification

Lecturer: Chitchanok Chuengsatiansup

Semester 2, 2024

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These slides include materials from:

Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development, Third Edition, by Craig Larman, Pearson Education Inc., 2005.